Initial Project Planning

Medical Report Analyzer — Course: CSE4204

1. Project Information

Project Title	Medical Report Analyzer
Course	CSE4204
Team Name	CSE4204-8D-T02 (Section 8D)
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2. Problem Statement

Medical reports are filled with clinical terminology and technical jargon that the average patient cannot understand. Patients receiving blood test results, lab reports, or discharge summaries are often left confused about what their findings mean and what actions to take next. Existing tools rely on generic search engines or rigid rule-based parsing that cannot handle the wide variety of report formats, abbreviations, and layouts used across different hospitals and labs. No mainstream tool offers a personalized, conversational way for a patient to interact with their own report. A large language model is uniquely suited to read unstructured medical text, extract what matters, and explain it in plain language.

3. Project Objectives

- Allow users to paste text or upload any medical report (PDF or image)
- Generate an instant plain-language AI summary with key findings highlighted
- Flag abnormal lab values or results for immediate user attention
- Provide an interactive Q&A chatbot grounded in the specific report content
- Save all reports, summaries, and chat history securely per user account
- Support scanned and image-based reports using OpenAI Vision
- Ensure private, per-user access through JWT-based authentication

4. Development Timeline

Phase	Deliverable	Timeline
1-Setup	Repository setup, Supabase configuration, base Express server initialization, React scaffolding.	Weeks 2- 3
2-Auth	User registration, login flows, JWT middleware injection, protected route configurations, and Auth UI components.	Week 4
3-Input	Text input pipelines, asynchronous file uploads targeting Supabase Storage, and active report persistence hooks in the database.	Weeks 5
4-AI	Open AI core summarization integrations, deep clinical entity extraction layers, and conditional data outlier flags.	Weeks 6-7
5-Chatbot	Context-bounded Q&A chatbot execution system with active prompt grounding; stateful chat logs committed to DB tables.	Weeks 8
6-Dashboard	Complete history rendering view, structured clean summary presentation panels, and final UI layout refinement.	Weeks 9
7-Deploy	Comprehensive systemic testing, validation passes, edge bug remedies, and absolute Vercel + Render host packaging.	Weeks 10

6. Expected Outcome

The completed system will allow any patient, regardless of medical background, to understand their own health report within seconds of uploading it. The AI chatbot will answer personalized questions, reducing unnecessary follow-up consultations. Secure per-user storage gives patients a personal health record they can revisit at any time. Vision AI removes the barrier of image-only reports, and the conversational interface creates an experience that no static tool currently provides.